DATA CLEANING USING SQL

```
-- database is created
CREATE DATABASE zomato_analysis;
USE zomato_analysis;

-- table is created and data is imported
CREATE TABLE zomato_cleaned (
name VARCHAR(255),
online_order VARCHAR(10),
book_table VARCHAR(10),
rate VARCHAR(20),
votes INT,
approx_cost_for_two_people INT,
listed_in_type VARCHAR(50)
);
SELECT * FROM zomato_cleaned;

-- Checked for NULL values in the table/dataset
SELECT
        SUM(name IS NULL) AS name_nulls,
  SUM(online_order IS NULL) AS online_order_nulls,
  SUM(book_table IS NULL) AS book_table_nulls,
  SUM(rate IS NULL) AS rate_nulls,
  SUM(votes IS NULL) AS votes_nulls,
  SUM(approx_cost_for_two_people IS NULL) AS cost_nulls,
  SUM(listed_in_type IS NULL) AS type_nulls
FROM zomato_cleaned;
```

-- Checked for DUPLICATES in the table/dataset

```
SELECT
        COUNT(*) AS total_rows,
        COUNT(DISTINCT CONCAT_WS('|',
                name,
    online_order,
    book_table,
    rate,
    votes,
    approx_cost_for_two_people,
    listed_in_type
        )) AS UNIQUE_ROWS
FROM zomato_cleaned;
```

-- So the rate column contains ratings stored as text, not numbers.

```
SELECT DISTINCT rate
FROM zomato_cleaned
ORDER BY rate;
```

-- Check for invalid ratings like NEW, -, or anything else that isn't in the number/5 format.

```
SELECT DISTINCT rate
FROM zomato_cleaned
WHERE rate NOT REGEXP '^[0-5]([.][0-9])? ?/5$';
```

-- Temporarily disable Safe Update Mode

```
SET SQL_SAFE_UPDATES = 0;
```

-- Clean the rate column and we need to remove /5 and the possible space.

```
UPDATE zomato_cleaned SET rate = TRIM(REPLACE(rate, '/5', ''));
```

-- Safe mode is turned ON.

```
SET SQL_SAFE_UPDATES = 1;
```

-- All the values are cleaned.

```
SELECT DISTINCT rate
FROM zomato_cleaned
ORDER BY rate;
```

-- Convert rate from VARCHAR to DECIMAL

```
ALTER TABLE zomato_cleaned
MODIFY COLUMN rate DECIMAL(3,1);
```

-- Check for values in online_order column.

```
SELECT DISTINCT online_order
FROM zomato_cleaned;
```

-- There are Yes or No values in it.

-- Check values in book_table column.

```
SELECT DISTINCT book_table
FROM zomato_cleaned;
```

-- There are Yes or No values in it.

-- Check values in listed_in_type column.

```
SELECT DISTINCT listed_in_type
FROM zomato_cleaned;
```

-- All are consistent values in the above three columns like online_order, book_table, and listed_in_type.

ANSWER BUSINESS QUESTIONS

-- 1) What types of restaurants are most common?

```
85      # ANSWER BUSINESS QUESTIONS
86      -- 1) What types of restaurants are most common?
87  •   SELECT
88          listed_in_type AS restaurant_type,
89          COUNT(*) AS restaurant_count
90      FROM zomato_cleaned
91      GROUP BY listed_in_type
92      ORDER BY restaurant_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

restaurant_type	restaurant_count
Dining	110
Cafes	23
other	8
Buffet	7

2) Which Restaurant Type receives the Most Votes?

```
94      -- 2. Which restaurant type receives the most votes?
95  •   SELECT
96          listed_in_type AS restaurant_type,
97          SUM(votes) AS total_votes
98      FROM zomato_cleaned
99      GROUP BY listed_in_type
100     ORDER BY total_votes DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

restaurant_type	total_votes
Dining	20363
other	9367
Cafes	6434
Buffet	3028

3) How many restaurants offer online ordering?

```sql
-- 3) How many restaurants offer online ordering?
SELECT
    online_order,
    COUNT(*) AS restaurant_count
FROM zomato_cleaned
GROUP BY online_order;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

online_order	restaurant_count
Yes	58
No	90

4) What is the rating distribution?

```sql
-- 4) What is the rating distribution?
SELECT
    rate,
    COUNT(*) AS restaurant_count
FROM zomato_cleaned
GROUP BY rate
ORDER BY rate;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

rate	restaurant_count
2.6	1
2.8	2
2.9	7
3.0	2
3.1	7
3.2	7
3.3	14
3.4	12
3.5	6
3.6	11
3.7	15
3.8	20
3.9	10

5) What price range is most common for two people?

```
117     -- 5) What price range is most common for two people?
118 •   SELECT
119         approx_cost_for_two_people AS common_cost,
120         COUNT(approx_cost_for_two_people) AS cost_count
121     FROM zomato_cleaned
122     GROUP BY approx_cost_for_two_people
123     ORDER BY cost_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

common_cost	cost_count
300	23
200	16
150	16
400	15
500	14
600	13
800	12
450	6
100	6
250	6
700	5
550	3
750	3

6) Do online-order restaurants have better ratings?(YES)

```
124
125     -- 6) Do online-order restaurants have better ratings?
126 •   SELECT
127         online_order,
128         ROUND(AVG(rate),2) AS average_rating
129     FROM zomato_cleaned
130     GROUP BY online_order;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

online_order	average_rating
Yes	3.86
No	3.49

7) Which restaurant type has more online orders?

```sql
-- 7) Which restaurant type has more online orders?
SELECT
    listed_in_type,
    COUNT(online_order) AS restaurant_count
FROM zomato_cleaned
GROUP BY listed_in_type
ORDER BY restaurant_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

listed_in_type	restaurant_count
Dining	110
Cafes	23
other	8
Buffet	7

8) Which restaurants have the highest votes?

```sql
-- 8) Which restaurants have the highest votes?
SELECT
    name,
    votes
FROM zomato_cleaned
ORDER BY votes DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	votes
Empire Restaurant	4884
Meghana Foods	4401
Onesta	2556
Onesta	2556
Kabab Magic	1720

9) What are the most popular restaurant types based on customer engagement?

```sql
-- 9) What are the most popular restaurant types based on customer engagement?
SELECT
    listed_in_type AS restaurant_type,
    COUNT(*) AS restaurant_count,
    SUM(votes) AS total_votes,
    ROUND(AVG(votes),2) AS avg_votes
FROM zomato_cleaned
GROUP BY listed_in_type
ORDER BY avg_votes DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

restaurant_type	restaurant_count	total_votes	avg_votes
other	8	9367	1170.88
Buffet	7	3028	432.57
Cafes	23	6434	279.74
Dining	110	20363	185.12